

Applied Programming for AI

Project description - June 2025

The main objective of this project is to develop a web application using a RESTful architectural style ¹. Your application must interact with external APIs to provide additional functionality or data. The development of your project must follow a Test-Driven Development (TDD) approach ². This project will be completed in groups of at most four students. All team members must contribute to the application's design and development. However, the grade is individual (see Canvas for more details).

¹ Roy Thomas Fielding and Richard N. Taylor. *Architectural styles and the design of network-based software architectures*. PhD thesis

² Kent Beck. *Test driven development: By example*. Addison-Wesley Professional, 2022

Project requirements

General remarks

- - Students must create a virtual environment for their projects (e.g., Anaconda, Virtual env.) ³
- Projects must have a `requirements.txt`. ⁴
- All functions and classes must have docstrings.
- All functions must have type hints for the parameters and return values.
- Each team member **must** create their own branch to work on for each feature they are working on.

³ For example, if you want to use virtual env you can use this command: `virtualenv .venv`.

⁴ To do so, you can use the following command: `pip freeze > requirements.txt`.

Team composition

Each team should consist of at most FOUR students. Ensure that roles and responsibilities are evenly distributed among team members.

Project's theme

CHOOSE A THEME OR DOMAIN for your application (e.g., travel, finance, health, education, social networking). The application should address a specific problem or provide a unique service within the chosen domain.

EXAMPLE: Data-self portrait⁵.

Create a visualization that presents insights from a dataset about yourself. You may use pre-existing data (such as your email archive, fitness tracker data, etc.) or create a new system specifically for collecting data about an aspect of your life. The data you collect need not be temporal; for example, you might reimagine your wardrobe as a database of interrelated items. Can you collect data about phenomena that nobody has seen or thought of before?

⁵ This is an adaption of

Golan Levin and Tega Brain. *Code as creative medium: a handbook for computational art and design*. MIT Press, 2021

Your visualization is a tool you're building to help you answer a question about yourself. You can use existing measurement technologies, or devise new manual or automatic data collection techniques. You're encouraged (but not required) to combine multiple sources of data, to make interesting comparisons.

Architecture

The system must include at least one web server to handle the application's backend and frontend operations. Keep in mind that the frontend and the backend must be split into different applications.

- Frontend: responsible for the interaction with the user, and visualization of the different services.
- Backend: responsible for implementing the logic and persistency of the application and communication with 3rd party services.

Communication Protocol

All communication between the frontend and backend shall occur via RESTful services to ensure standardized and efficient data exchange.

Collaborative development

VERSION CONTROL. Manage all code changes using Git to ensure version control and collaboration among team members.

Backend Server Functionality

This server implements the logic and persistence layer of your application. You must utilize the Python programming language for developing this server. Your application must contain at least the following functionalities⁶ ⁷.

To implement the backend you will follow a TTD approach. Therefore, you will follow the next process:

1. Implement the test(s) for the specific feature.
2. Commit the test once you finish the test definition.
3. Implement the desired feature, until all tests for this feature pass.
4. Refactor your code.

Data structures: You must define your own data structures to model the application and the chosen domain.

⁶ Remember that this are guidelines, your own project might and should deviate from them based on your discussions with your stakeholder.

⁷ When grading, we will take into account the complexity of the implemented features—meeting the minimum requirements will earn a passing score, but going beyond can boost your grade.

CRUD Operations: Users shall be able to perform Create, Read, Update, and Delete operations on relevant data entities within the application ⁸.

Persistence : The application must utilize persistent storage mechanisms for data management, and it should allow users to load existing data.

Query: The application must offer a query service with a minimum of three different filters to allow users to refine their search queries effectively.

Data cleaning: The backend API must enrich the data retrieved from external sources before sending it to the frontend.

⁸ For example, the [FeastFinder](#) project, the app must allow users to perform CRUD operations on recipes.

Frontend Functionality

The frontend implements the user interface of your application. You must develop this "individually" ⁹ using HTML, JavaScript, and CSS. Existing frameworks can be leveraged for efficiency. Your application must contain at least the following functionalities¹⁰:

⁹ Remember that for the implementation of the frontend you can collaborate with generative AI tools for coding.

¹⁰ Remember that these are only guidelines; your own project might deviate from them based on your discussions with your stakeholder.

Landing Page: Upon accessing the application, users shall be directed to a visually appealing landing page providing an overview of the application's purpose and features.

CRUD Operations: Users shall be able to perform Create, Read, Update, and Delete operations on relevant data entities within the application.

Search Page: The application must feature a search page equipped with a minimum of three different filters to allow users to refine their search queries effectively.

Detail Result Page: Upon selecting an item from the search results, users shall be directed to a detailed result page displaying comprehensive information about the selected item.

CRUD Operations: Users shall be able to perform Create, Read, Update, and Delete operations on relevant data entities within the application. For example, in the Project Example (@Márton Bodó let's include a link to the project), the app must allow users to perform CRUD operations on recipes.

Visualization: The application shall incorporate visualization components to present data in an intuitive and visually appealing manner, enhancing user understanding and engagement. Consider how your project can support different operations such as zooming, and sorting.

UI Design Standards

Consistent Template: Maintain consistency throughout the UI design by avoiding collage-like layouts.

User Interface Feedback: The user interface (UI) must provide timely and clear feedback to users for their interactions with the application, ensuring a smooth user experience. For example, handling errors when something goes wrong.

Responsiveness: Develop a responsive user interface that adapts to various screen sizes and devices.

UI Alignment: Ensure proper alignment of UI elements for a visually pleasing presentation.

Color Palette: Utilize a proper color palette, such as one generated from Adobe Color, to maintain visual harmony and consistency.

Deliverables

There are different deliverables for this project. For more details and specific deadlines please check Canvas.

- Project proposal
- Backend deliverable including tests ¹¹.
- Frontend deliverable ¹².
- Final demo presentation
- Peer-review
- Generative AI Report
- Coding interview

¹¹ This will take place as a release in the repository.

¹² This will take place as a release in the repository.

References

Kent Beck. *Test driven development: By example*. Addison-Wesley Professional, 2022.

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